

***LEGO** PIECE FINDER*

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Context

LEGO is the world's most popular building brick, enjoyed by children and adults everywhere. However, once the bricks have been mixed up, it is difficult to tell which set they originally belong to. It takes a long time to do this by hand, especially with large collections.

The objective of this project is to create an AI system that will use image recognition to identify LEGO bricks and match them with their corresponding LEGO set. Leveraging the capabilities of a Kaggle dataset containing images of LEGO bricks and sets, the system will help LEGO enthusiasts, collectors, and children sort their collections more efficiently.

Scope

This project seeks to create an AI system capable of identifying LEGO bricks and determining the sets that they are a part of. Utilizing a dataset of LEGO brick images from Kaggle, a model will be learned to classify single parts. Identified parts will then be correlated with sets via a LEGO parts database.

The project will focus on the development and testing of a prototype model and exploration of a simple-to-use interface for uploading a brick image and receiving predictions. Industrial deployment or commercial application is out of scope; it is intended to demonstrate precise part detection and set identification.

How does it benefit?

Such a system can save time, reduce frustration, and allow families, collectors, and resellers to better organize their LEGO collection. It also encourages sustainability by reuse and reduction of plastic waste.

In the future, the model could be amplified to recognize sets of bricks, and it would be more precise and have a greater likelihood of pairing them with their respective sets. With ongoing technological innovation, it could be integrated into applications or databases that would aid in rebuilding, aware of pieces that were missing, and even suggest replacements creating a more efficient LEGO system.

Technical Approach

For image recognition, the project will use YOLO, a proven real-time object detection model well-suited for identifying small, distinct items like LEGO pieces. Training will be based on a Kaggle dataset of individual LEGO parts, which will then be linked to a complementary Kaggle dataset of LEGO sets. This combination allows the system to classify bricks accurately and match them with their corresponding sets.

Learning Outcomes

Societal Impact

This project improves the LEGO hobbyist experience by making set identification and organization faster and easier. It encourages reuse of bricks, reduces waste, and helps collectors manage large collections. In the long term, expanding the model to recognize groups of bricks could further improve accuracy and enable integration with apps or databases, supporting rebuilding, replacement, and a more sustainable LEGO ecosystem.

Investigative Problem Solving

Challenges include handling visually similar bricks, dataset imbalances, and potential mistakes. Solutions such as data augmentation, processing techniques, and model fine-tuning will be explored.

Data Preparation

The image dataset will be cleaned, normalized, and augmented, while the LEGO database will be prepared to allow efficient mapping of parts to sets. Any dataset limitations, such as missing bricks or rare parts, will be documented.

Machine Learning

The model will be tested and trained to guarantee that it generalizes correctly to new images. Model performance will be quantified to allow for improvement and determine readiness for use in a prototype.

Data Visualization

Visualizations such as algorithm performances, instances in different categories, and examples of correct and incorrect predictions will be used to communicate results effectively to both technical and non-technical audiences.

Reporting

The project will be documented using Jupyter Notebooks and structured work reports. The notebooks will include code, visualizations, and annotations to ensure transparency and reproducibility. Work reports will summarize progress, key decisions, and challenges, making the results accessible to both contextual and technical stakeholders.

Personal Leadership

This project expands my technical skills in AI and computer vision and expands my autonomy and ownership too. I want to grow by carrying out the whole process independently, from research and dataset preparation to model training and testing. By controlling each step, I hope to gain problem-solving confidence, decision-making, and applying technical knowledge to a real-world problem and hope I can validate this to my coach and consultants. This will allow me to be more self-sufficient as well as further my understanding in designing end-to-end AI solutions.

Personal Goal (Learning to Work Independently)

My goal for this project is to challenge myself to improve how I organize my own work and manage time effectively. I want to develop the ability to plan and execute tasks independently, stay disciplined, and structure my schedule in a way that ensures steady progress. By taking ownership of my workflow, I aim to make the personal challenge a success and grow my self-management skills for future projects.